Finite Groups and Their Representations — Exercise Sheet 2

Quotients and products

Variant: Questions only

Exercise 1 (A quotient of a quotient). Let $N \leq M$ be normal subgroups of a finite group G.

(i) Prove the third isomorphism theorem: $M/N \trianglelefteq G/N$ and $(G/N)/(M/N) \cong G/M$. Write the isomorphism explicitly.

(ii) Take $G = C_{24} = \langle \zeta \rangle$, $N = \langle \zeta^{12} \rangle$ and $M = \langle \zeta^4 \rangle$. Identify all three quotient groups in the first part.

Exercise 2 (The determinant quotient). Let $\mathbb{F}_3 = \{0, 1, 2\}$ be the field with arithmetic modulo 3. Let $G = \mathrm{GL}_2(\mathbb{F}_3)$ be the group of invertible 2×2 matrices over $\mathbb{F}_3$, and put $N = \mathrm{SL}_2(\mathbb{F}_3) = \{A \in G : \det A = 1\}$. Write $\mathbb{F}_3^\times = \{1, 2\}$ for the group under multiplication modulo 3.

(i) Use the determinant to identify G/N. Find $|G|$ and $|N|$.

(ii) Put $h = \mathrm{diag}(2, 1)$ and $H = \langle h \rangle$. Show that $G = N \rtimes H$. Describe the action of H on N.

Exercise 3 (External semidirect products). Let N, H be finite groups. An *automorphism* of N is a bijective homomorphism $N \to N$. Write $\mathrm{Aut}(N)$ for the group of these maps under composition. Let $\varphi \colon H \to \mathrm{Aut}(N)$ be a homomorphism. On $N \times H$ consider the multiplication

$$(n, h)(n', h') = (n\varphi(h)(n'), hh').$$

(i) Prove that this is a group, written $N \rtimes_\varphi H$. Find its identity and inverses and identify its copies of N and H.

(ii) For $m \geq 2$, choose a generator x of C_m. Write $(\mathbb{Z}/m\mathbb{Z})^\times$ for the residue classes u modulo m with $\gcd(u, m) = 1$, under multiplication modulo m. You may use Bézout's identity: if $\gcd(u, m) = 1$, there are integers p, q with $pu + qm = 1$. Prove $\mathrm{Aut}(C_m) \cong (\mathbb{Z}/m\mathbb{Z})^\times$. Deduce $\mathrm{Aut}(C_8) \cong V_4$.

(iii) Let $N = \langle x \rangle \cong C_8$ and $H = \langle y \rangle \cong C_2$, with $\varphi(y)(x) = x^3$. Show that this gives a nonabelian group of order 16, and find its centre.

Exercise 4 (A dihedral presentation). For $n \geq 3$, let P be the group presented by generators u, v and relations $u^n = v^2 = e$, $vuv^{-1} = u^{-1}$. Its elements are words in u, v and their inverses, modulo the smallest equivalence compatible with products and inverses that imposes these relations and the inverse cancellations. Prove $P \cong D_n$. Your argument must show both that there are at most $2n$ elements and that no further identifications between the normal forms occur.

Exercise 5 (A permutation and its power). In S_9, let $\sigma = (1\,2\,3\,4)(5\,6\,7)(8\,9)$. Find its order, sign, centraliser order and class size. Find the cycle type, centraliser order and class size of σ^6. Finally give a permutation τ with $\tau\sigma\tau^{-1} = \sigma^{-1}$.

Exercise 6 (Uniqueness of sign). For $n \geq 2$, prove that the only homomorphisms $S_n \to \{1, -1\}$ are the trivial map and sgn. More generally, if A is abelian, show that a homomorphism $f \colon S_n \to A$ is determined by a single element $t \in A$ with $t^2 = e$, and that every such t gives a homomorphism. State what changes when $n = 1$.

Exercise 7 ⋆. Put

$$a = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}, \qquad b = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Q_8 = \{\mathbb{1}, -\mathbb{1}, a, -a, b, -b, ab, -ab\} \subset \mathrm{GL}_2(\mathbb{C}).$$

An *involution* is an element of order 2. Show that Q_8 is a group and that it has a unique involution. Deduce that Q_8 cannot be an internal semidirect product of two proper subgroups. Identify $Q_8/\{\mathbb{1}, -\mathbb{1}\}$.